AI Agent, GenAI, LLM & ML Interview Preparation (Backend Engineer)

1. Tell me about your AI experience

I have worked on integrating LLM APIs into Python/FastAPI applications, building AI-powered APIs, prompt engineering, document processing, RAG workflows, vector databases, and automation using n8n. My focus has been backend integration rather than training foundation models.

2. AI vs ML vs DL vs GenAI

AI: Intelligent systems. ML: Learns from data. Deep Learning: Neural networks. GenAI: Generates text, images, code, etc.

3. LLM

Large Language Models are trained on huge datasets to understand and generate human language. Examples: GPT, Gemini, Claude, Llama.

4. Prompt Engineering

Design prompts with Role, Task, Context, Constraints and Output format.

5. Temperature & Top-P

Temperature controls randomness. Low for coding, high for creativity. Top-P limits token selection by probability.

6. Tokens

Models process tokens instead of words. Token usage affects context window and cost.

7. Embeddings

Numerical vector representations of text used for semantic search.

8. Vector Database

Stores embeddings. Examples: FAISS, ChromaDB, Pinecone, Weaviate, Milvus.

9. RAG

Flow: User -> Embedding -> Vector Search -> Relevant Chunks -> LLM -> Answer. Reduces hallucinations.

10. Hallucination

Incorrect but confident answers from LLMs. Mitigate using RAG, good prompts, and citations.

11. Fine-tuning vs Prompt Engineering

Prompt engineering changes instructions. Fine-tuning retrains the model on custom data.

12. AI Agent

Uses an LLM plus memory, planning, tool calling, reasoning and execution to complete multi-step tasks.

13. Function Calling

LLM decides to invoke external APIs (calendar, email, DB, search) to perform actions.

14. MCP

Model Context Protocol is a standard for connecting AI models to tools and data sources.

15. n8n

Popular automation platform to orchestrate AI workflows using HTTP, Gmail, Drive, Slack, databases and LLMs.

16. ML Basics

Training, validation, testing; metrics: Accuracy, Precision, Recall, F1, MAE, RMSE.

17. FastAPI AI Architecture

User -> FastAPI -> Chunking -> Embeddings -> Vector DB -> Retrieve Top-K -> LLM -> Response.

18. Frequently Asked Questions

• What is RAG? • What are embeddings? • Explain vector databases. • Difference between AI Agent and Chatbot. • What is prompt engineering? • Explain hallucination. • What is function calling? • How would you scale an AI API? • How do you secure LLM applications? • Explain async FastAPI.

19. Backend Topics

JWT, Redis, PostgreSQL, Docker, AsyncIO, Streaming responses, Background tasks, Rate limiting, Logging.

20. Tips

Be honest about your responsibilities. Emphasize backend integration, APIs, architecture, and problem solving instead of claiming model training experience.